Week 3 | **CLO** 2-A · 3-C | **Textbook** p. 178 {2, 3, 10, 17} | **File** 3 / 8

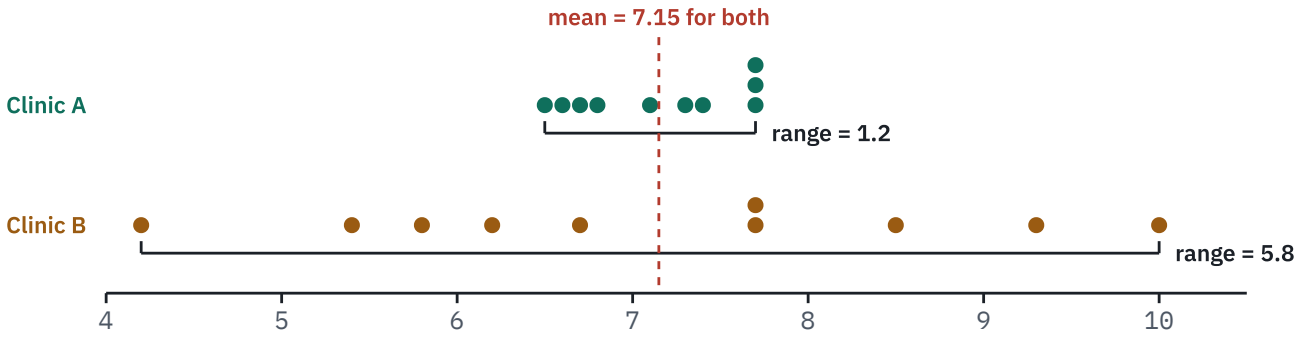
3-2 Measures of Variation

مقاييس التشتت — المدى والتباين والانحراف المعياري ومعامل الاختلاف ونظرية تشيبيشيف

Variation (spread, dispersion) describes how far the data values are from each other, or how dispersed they are. The main measures are the **range**, **variance**, **standard deviation** and **interquartile range** (covered in 3-3). A measure of variation combined with a measure of centre gives a good picture of a distribution. BOTH

0 · Why centre is not enough AMANI

Waiting times (minutes) at two SQU hospital clinics. A: 6.5 6.6 6.7 6.8 7.1 7.3 7.4 7.7 7.7 7.7 · B: 4.2 5.4 5.8 6.2 6.7 7.7 7.7 8.5 9.3 10.0. Both clinics have mean 7.15, median 7.2, mode 7.7 and midrange 7.1, so the centre cannot separate them. The dot plot shows the difference.



Same mean, median, mode and midrange, but clinic B's waiting times are spread almost five times wider. The centre alone cannot tell the clinics apart; a measure of variation can.

نفس المتوسط والوسيط والمنوال ومنتصف المدى للعيادتين، لكن أوقات الانتظار في العيادة B مشتّتة أكثر بكثير. مقاييس النزعة المركزية وحدها لا تكفي، ونحتاج مقاييس التشتت.

1 · The range BOTH

RANGE

$$R = \text{highest value} - \text{lowest value}$$

The simplest measure, but it depends only on the two extremes, so one extreme value changes it markedly. It is less useful than the variance and SD. Give it as a **single number**.

SIMPLE EXAMPLE AMANI · **EX 3-16 PAINT** HADHRAMI

(a) 1, 2, 4 · (b) 1, 2, 1000 · (c) months before fading: brand A 10 60 50 30 40 20, brand B 35 45 30 35 40 25.

(a) R = **3** · (b) R = **999**: one extreme value multiplies the range by 333. (c) A: `60 − 10 =` **`50 months`**; B: `45 − 25 =` **`20 months`**, less than half of brand A's. Textbook: Ex 3-15, 3-17.

2 · Variance and standard deviation BOTH

Both are built on the **deviation**, the distance of each value from the mean: $(X - \mu)$ or $(X - \bar{x})$. Deviations always sum to zero, $\Sigma(X - \bar{x}) = 0$, so they are squared before averaging. The **variance** is the average of the squared deviations, and the **standard deviation** is its square root.

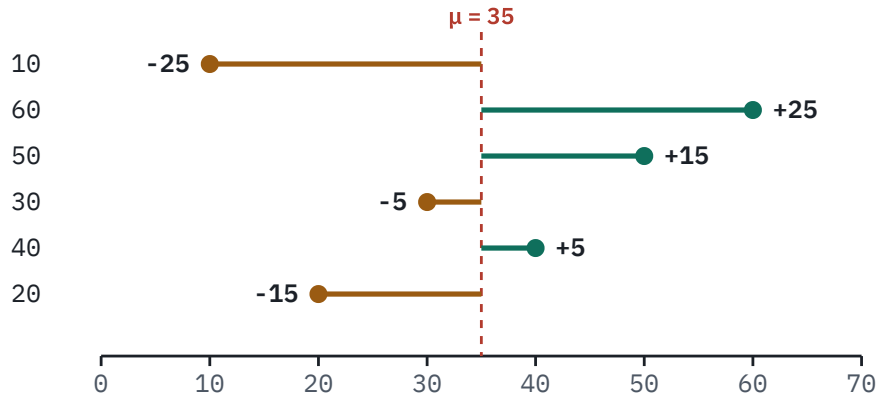
	POPULATION (PARAMETER)	SAMPLE (STATISTIC)
Variance التباين	$\sigma^2 = \frac{\Sigma(X - \mu)^2}{N}$	$s^2 = \frac{\Sigma(X - \bar{x})^2}{n - 1}$
Standard deviation الانحراف المعياري	$\sigma = \sqrt{\sigma^2}$	$s = \sqrt{s^2}$
Shortcut (sample)	$s^2 = \frac{n(\Sigma X^2) - (\Sigma X)^2}{n(n - 1)}$	or $s^2 = \frac{\Sigma X^2 - n \cdot \bar{x}^2}{n - 1}$

PROPERTIES TO QUOTE

Variance and SD can **never be negative**, so take the positive square root. If the values sit close to the mean, the variance is small. The **serious disadvantage of the variance** is that its units are squared (units²). The **SD has the same units as the data**. The larger the SD, the more spread out the data. When two data sets have equal means, the one with the larger variance or SD is more variable. ΣX^2 (square, then sum) is not $(\Sigma X)^2$ (sum, then square).

RESEARCH QUESTION IN BOTH DECKS: WHY N − 1?

The sample mean $\bar{x}$ is computed from the same data, so the values sit closer to $\bar{x}$ than to the true μ. Dividing by n would therefore **underestimate** σ^2. Dividing by $n - 1$ (the *degrees of freedom*: once $\bar{x}$ is fixed, only n − 1 deviations are free) gives a slightly larger value, which makes s^2 an **unbiased** estimator of σ^2.



$\Sigma(X - \mu) = 0$

squares:
625+625+225+25+25+225

$\Sigma(X - \mu)^2 = 1750$

$\sigma^2 = 1750 / 6 \approx 291.7$

$\sigma = \sqrt{291.7} \approx 17.1$

Example 3-18 (brand A paint). Each bar is a deviation X − μ. They cancel to zero, which is why we square them before averaging; the average squared deviation is the variance.

مثال 3-18: كل خط يمثل انحراف القيمة عن المتوسط. مجموع الانحرافات = صفر دائمًا، لذلك نربّعها قبل أخذ المتوسط، ومتوسط مربعات الانحرافات هو التباين.

EX 3-18 · PAINT BRAND A (A POPULATION) HADHRAMI

10, 60, 50, 30, 40, 20. Find σ^2 and σ.

μ = 210/6 = 35; $\Sigma(X - \mu)^2 = 1750$ (see the figure).

$\sigma^2 = 1750/6 =$ **291.7** $\sigma =$ **17.1 months**

Brand B (textbook Ex 3-19): μ = 35, $\Sigma(X - \mu)^2 = 250$, so $\sigma^2 = 41.7$ and $\sigma = 6.5$. Same mean, but brand A is far more variable.

SIMPLE EXAMPLE: 1, 2, 4 (SAMPLE) AMANI

Find s^2 and s.

$\bar{x} = 7/3$. Deviations: −4/3, −1/3, 5/3, which sum to 0. Squares: 16/9 + 1/9 + 25/9 = 42/9 = 4.667.

$s^2 = 4.667/2 =$ **2.333** $s =$ **1.53**

The notes print 2.335 because the deviations were rounded to −1.33, −0.33, 1.67 first. That is exactly why the rounding rule says **never round mid-calculation**.

EX 3-20 AND 3-21 · TEACHER STRIKES HADHRAMI = EJADA TEACHER AWARDS AMANI

A random sample of years: 9, 10, 14, 7, 8, 3. Find s^2 and s by the definition formula and by the shortcut formula.

Definition: $\bar{x} = 51/6 = 8.5$. Deviations 0.5, 1.5, 5.5, −1.5, −0.5, −5.5; squares 0.25, 2.25, 30.25, 2.25, 0.25, 30.25.

$s^2 = 65.5/5 = 13.1$

Shortcut: $\Sigma X = 51$, $\Sigma X^2 = 499$.

$$s^2 = \frac{6(499) - 51^2}{6(5)} = \frac{2994 - 2601}{30} = \frac{393}{30} = 13.1$$

Check: $(499 - 6 \cdot 8.5^2)/5 = 65.5/5 = 13.1$

All three formulas agree: **$s^2 = 13.1$, $s = \sqrt{13.1} \approx 3.6$**

ACTIVITY · RANGE, VARIANCE, SD OF BOTH CLINICS AMANI

CLINIC A X	X − 7.15	(X − 7.15)²
6.5	−0.65	0.4225
6.6	−0.55	0.3025
6.7	−0.45	0.2025
6.8	−0.35	0.1225
7.1	−0.05	0.0025
7.3	0.15	0.0225
7.4	0.25	0.0625
7.7 ×3	0.55	0.3025 ×3
Σ	0	**2.045**

A: R = `1.2`, $s^2 = 2.045/9 =$ **0.227**, s = **0.477 min**.

B: R = `5.8`, $\Sigma(X - \bar{x})^2 = 29.865$, $s^2 =$ **3.318**, s = **1.822 min**.

Clinic B's SD is almost **four times** clinic A's. A is the more **consistent** clinic: patients there know what wait to expect.

ACTIVITY · CAR ACCIDENTS IN MUSCAT AMANI

Five months: 22, 30, 36, 24, 5 (note: mean = 28). Find R, s^2, s.

Version consistent with the note (last value 28): $\bar{x} = 140/5 = 28$; deviations −6, 2, 8, −4, 0; Σ = 120.

$R = 36 - 22 =$ **14** $\quad s^2 = 120/4 =$ **30** $\quad s =$ **5.48**

As printed (last value 5): $\bar{x} = 23.4$, R = 31, $\Sigma(X - \bar{x})^2 = 543.2$, $s^2 = 135.8$, s = 11.65. One low month nearly doubles the SD.

الشرح بالعربي — التباين والانحراف المعياري

الانحراف = بُعد القيمة عن المتوسط $X - \bar{x}$. مجموع الانحرافات = صفر دائمًا، لذلك **نربّعها** ثم نأخذ متوسطها، وهذا هو التباين. التباين وحدته **مربعة** (وهذا عيبه)، فنأخذ الجذر لنحصل على الانحراف المعياري، وهو بنفس وحدة البيانات. للمجتمع نقسم على N، وللعينة نقسم على $n - 1$. انتبه: اقرأ السؤال جيدًا. «عينة» تعني $n - 1$، و«مجتمع» أو «كل الموظفين» تعني N. والقانون المختصر يعطي نفس الناتج تمامًا، لكن احذر الخلط بين ΣX^2 و $(\Sigma X)^2$.

3 · Variance and SD for grouped data BOTH

SHORTCUT FORMULA: TABLE A CLASS · B F · C X_m · D $F \cdot X_m$ · E $F \cdot X_m^2$

$$s^2 = \frac{n(\Sigma f \cdot X_m^2) - (\Sigma f \cdot X_m)^2}{n(n-1)} \qquad s = \sqrt{s^2}$$

Steps: find the midpoints; $f \cdot X_m$ → column D; $f \cdot X_m^2$ → column E; sum columns B (= n), D and E; substitute; take the square root.

EX 3-22 · MILES RUN PER WEEK (20 RUNNERS) HADHRAMI

CLASS	F	X_m	$F \cdot X_m$	$F \cdot X_m^2$
5.5–10.5	1	8	8	64
10.5–15.5	2	13	26	338
15.5–20.5	3	18	54	972
20.5–25.5	5	23	115	2,645
25.5–30.5	4	28	112	3,136
30.5–35.5	3	33	99	3,267
35.5–40.5	2	38	76	2,888
Σ	**20**		**490**	**13,310**

$$s^2 = \frac{20(13{,}310) - 490^2}{20(19)} = \frac{26{,}100}{380} \approx \boxed{68.7} \qquad s \approx \boxed{8.3 \text{ miles}}$$

AGES OF DRIVERS (TABLE IN 3-1) AMANI

Find the sample variance and SD.

Column E ($f \cdot X_m^2$): 11,404.75 · 21,125 · 26,244 · 32,931.5 · 0 · 4,160.25 · 5,256.25.

$n = 71$, $\Sigma f \cdot X_m = 2579.5$, $\Sigma f \cdot X_m^2 = 101{,}121.75$

$$s^2 = \frac{71(101{,}121.75) - 2579.5^2}{71(70)} = \frac{525{,}824}{4970} = \boxed{105.80}$$

$s = \boxed{10.29 \text{ years}}$

With $\bar{x} = 36.33$ from 3-1, the CV is 10.29/36.33 = 28.3%.

4 · Coefficient of variation (CVar) BOTH

COEFFICIENT OF VARIATION: SD AS A PERCENTAGE OF THE MEAN

$$\text{sample: } CVar = \frac{s}{\bar{x}} \cdot 100$$

$$\text{population: } CVar = \frac{\sigma}{\mu} \cdot 100$$

The SD and the mean share the same units, so CVar has **no units**. That is why it can compare standard deviations measured in different units, or data sets with different means. Lower CV: values closer together, **less variable and more consistent**. Higher CV: more spread out, greater relative variability.

EX 3-23 · CAR SALES VS COMMISSIONS BOTH

Sales: mean 87, s = 5. Commissions: mean 5225 OMR, s = 773.

sales $5/87 \cdot 100 \approx \boxed{5.7\%}$

commissions $773/5225 \cdot 100 \approx \boxed{14.8\%}$

The commissions are more variable than the sales.

EX 3-24 · ROLLER COASTERS HADHRAMI

Speed: mean 69.16 mph, **variance** 2.76. Height: mean 177.80 ft, **variance** 157.70.

ACTIVITY · TWO DEPARTMENTS AMANI

Dept 1: 40 employees, mean 450 OMR, SD 7. Dept 2: 65 employees, mean 350 OMR, SD 9. (1) Which has the higher total salary expense? (2) Which has greater variability?

Take the square root first, because the variance is given, not the SD:

$\sqrt{2.76/69.16} \cdot 100 \approx$ **2.4%**

$\sqrt{157.70/177.80} \cdot 100 \approx$ **7.1%**

The speeds are less variable and more consistent.

(1) The total is $n\cdot\bar{x}$: 40 × 450 = 18,000 and 65 × 350 = 22,750, so **Dept 2**.

(2) $CV_1 = 7/450 = 1.56\%$ and $CV_2 = 9/350 = 2.57\%$, so **Dept 2** varies more. Comparing raw SDs alone is not valid here because the means differ.

ACTIVITY · CV TO SD AMANI

80 candidates: mean height 145.8 cm, coefficient of variation 2.5%. Find the SD.

Rearrange the formula: `s = CV × x̄ / 100 = 0.025 × 145.8 =` **3.645 cm**. The 80 candidates is a distractor: n is not needed.

الشرح بالعربي — معامل الاختلاف

معامل الاختلاف = الانحراف المعياري ÷ المتوسط × 100، وهو **بدون وحدة**. نستخدمه للمقارنة بين مجموعتين **تختلف وحداتهما** (مثلًا عدد سيارات مقابل ريالات) أو **تختلف متوسطاتهما**. الأصغر يعني تشتتًا نسبيًا أقل واتساقًا أكبر. فخّان شائعان: (1) إذا أُعطيت **التباين** فخذ الجذر أولًا قبل القسمة. (2) لا تقارن الانحرافين المعياريين مباشرة إذا اختلف المتوسطان، بل قارن معاملي الاختلاف.

5 · Range rule of thumb BOTH

QUICK ESTIMATE OF s, ONLY FOR UNIMODAL AND ROUGHLY SYMMETRIC DATA

$$s \approx \frac{\text{range}}{4}$$

Why 4? (research question) For bell-shaped data about 95% of values lie within $\bar{x} \pm 2s$, an interval of width 4s, so the range is roughly 4s. The textbook uses the same idea to approximate the smallest and largest values: about $\bar{x} - 2s$ and $\bar{x} + 2s$.

EXAMPLES AMANI · HADHRAMI

(a) 12, 12, 14, 15, 16, 18, 18, 20, 20, 25. (b) 5, 8, 8, 9, 10, 12, 13.

(a) s ≈ 13/4 = 3.25, while the formula gives s = 4.06. (b) s ≈ 8/4 = 2, while the formula gives 2.7.

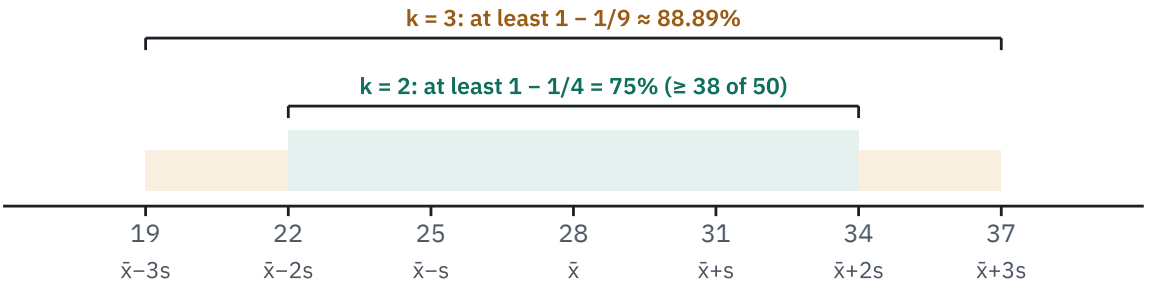
The rule is quick but less accurate. In most cases it **underestimates** s, but not always.

6 · Chebyshev's theorem: any shape BOTH

CHEBYSHEV (1821–1894): FOR ANY K > 1, NOT NECESSARILY AN INTEGER

$$\text{at least } 1 - \frac{1}{k^2} \text{ of the data lie within } \bar{x} \pm k \cdot s \text{ (sample) or } \mu \pm k \cdot \sigma \text{ (population)}$$

k = 1.5: at least 55.56% · k = 2: at least **75%** · k = 2.5: at least 84% · k = 3: at least **88.89%** (8/9). Finding k: `k = (upper limit − mean)/SD`. The answer is always "**at least**", never "exactly".



Dr. Amani's activity: n = 50, x̄ = 28, s = 3. The interval (22, 34) is x̄ ± 2s, so at least 75% (37.5, so at least 38 observations) lie inside, whatever the shape of the data.

نشاط د. أماني: n = 50 و x̄ = 28 و s = 3. الفترة (22, 34) هي x̄ ± 2s، إذن على الأقل من 75% من البيانات داخلها (أي 38 مشاهدة على الأقل) مهما كان شكل التوزيع.

ACTIVITY · N = 50, X̄ = 28, S = 3 AMANI

What can be said about the number of observations (1) inside (22, 34) and (2) outside it?

28 − 3k = 22, so k = (28 − 22)/3 = 2, and 1 − 1/4 = 75%.

(1) 0.75 × 50 = 37.5: **at least 38** observations inside. Round **up** for "at least".

(2) **at most 12** outside (50 − 38).

EX 3-26 · TRAVEL ALLOWANCE BOTH

Mean \$0.25/mile, s = \$0.02. Find the minimum percentage between \$0.20 and \$0.30.

k = (0.30 − 0.25)/0.02 = 2.5 → 1 − 1/6.25 = **84%**

EX 3-25 · HOUSE PRICES HADHRAMI

Mean \$50,000, s = \$10,000. Find the range containing at least 75% of prices.

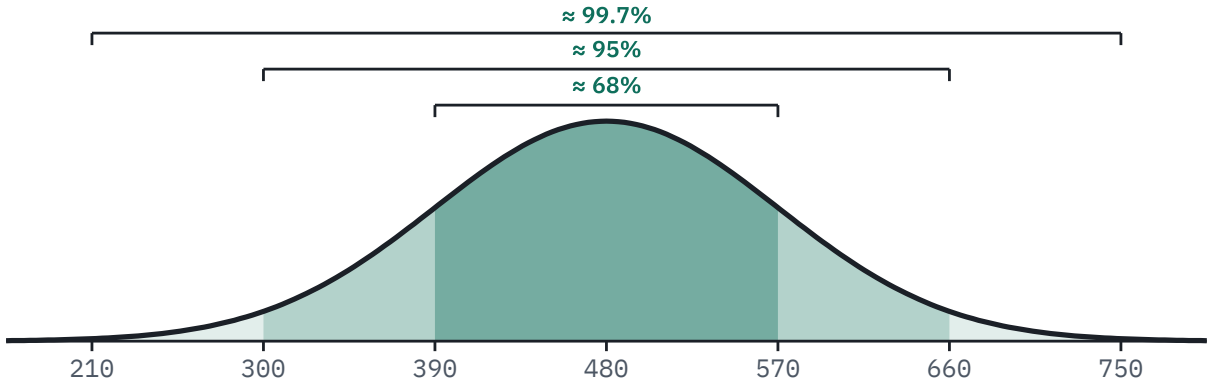
k = 2: 50,000 ± 20,000 → **\$30,000 to \$70,000**

7 · The empirical (normal) rule: bell-shaped only HADHRAMI

When the distribution is **bell-shaped (normal)**, approximately **68%** of values lie within 1 SD of the mean, **95%** within 2 SD, and **99.7%** within 3 SD. Chapter 6 develops this further.

	CHEBYSHEV	EMPIRICAL
Applies to	**any** shape	**bell-shaped** only
Within 2 SD	at least 75%	about 95%
Within 3 SD	at least 88.89%	about 99.7%
Wording	"at least"	"approximately"

Example: national exam scores with mean 480 and SD 90, normally distributed. About 68% lie between 390 and 570 (480 ± 90), 95% between 300 and 660 (480 ± 180), and 99.7% between 210 and 750 (480 ± 270).



The empirical rule holds only for bell-shaped (normal) data. Exam scores with mean 480 and SD 90: about 68% fall in 390–570, 95% in 300–660, 99.7% in 210–750.

القاعدة التجريبية تصلح فقط للبيانات الجرسية (الطبيعية). درجات بمتوسط 480 وانحراف 90: حوالي 68% بين 390 و 570، و95% بين 300 و 660، و99.7% بين 210 و 750.

الشرح بالعربي — تشيبيشيف والقاعدة التجريبية

نظرية تشيبيشيف تصلح لأي بيانات مهما كان شكلها، وتقول: **على الأقل** $1 - 1/k^2$ من البيانات تقع داخل $\bar{x} \pm k \cdot s$. الخطوة الأولى دائمًا: احسب $k =$ (الحد − المتوسط) ÷ الانحراف. ثم إذا طُلب **عدد** المشاهدات اضرب النسبة في n وقرّب **لأعلى** لـ«على الأقل». وما بالخارج = **على الأكثر** الباقي. أما القاعدة التجريبية (68 – 95 – 99.7) فتصلح **فقط** للتوزيع الجرسي (الطبيعي). إذا ذكر السؤال «bell-shaped» أو «normal» استخدم التجريبية، وإلا فاستخدم تشيبيشيف.

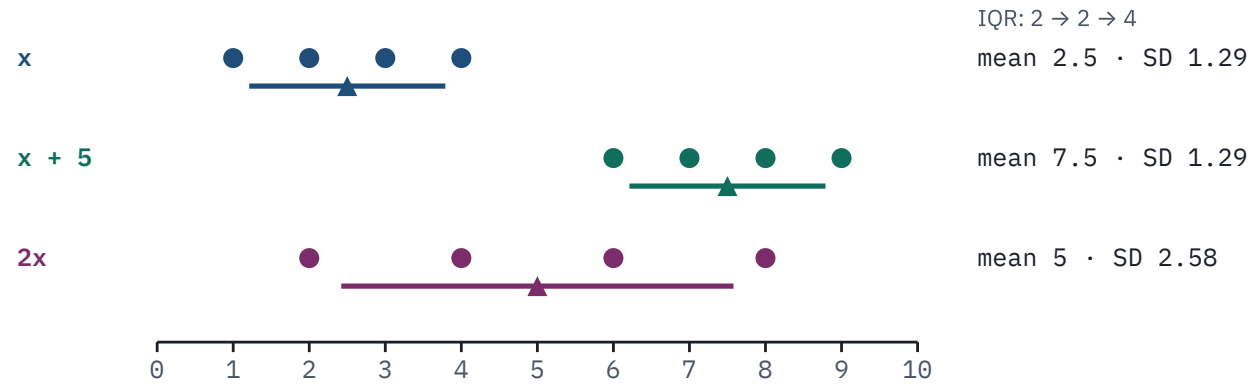
8 · Uses of variance and SD · Table 3-2 BOTH

(1) They describe the **spread** of data and compare which of two or more data sets is more variable. (2) They measure the **consistency** of a variable: in manufacturing nuts and bolts, the variation in diameters must be small or the parts will not fit. (3) They give the number of values that fall within an interval (Chebyshev). (4) They are used heavily in **inferential statistics** in later chapters.

MEASURE	DEFINITION	SYMBOL
Range المدى	highest – lowest	R
Variance التباين	average squared distance from the mean	σ^2, s^2

MEASURE	DEFINITION	SYMBOL
Standard deviation الانحراف المعياري	square root of the variance	σ, s

9 · Linear transformations HADHRAMI



Adding 5 slides every point (and the mean, median) right by 5 but leaves the spread (SD, IQR) unchanged. Multiplying by 2 doubles both the centre and the spread. Triangle = mean; bar = mean ± 1 SD.

إضافة 5 تُزيح كل القيم (والمتوسط والوسيط) بمقدار 5 لكن التشتت (SD و IQR) لا يتغير. الضرب في 2 يضاعف مقاييس المركز والتشتت معًا. المثلث = المتوسط، والخط = المتوسط ± انحراف معياري واحد.

TRANSFORMATION	MEAN, MEDIAN (CENTRE)	SD, IQR, RANGE (SPREAD)	VARIANCE	CV
Add a constant (x + c)	shift by + c	**unchanged**	unchanged	**changes** (the mean changed)
Multiply by a constant (c·x, c > 0)	× c	× c	**× c²**	unchanged
Both (a·x + b)	a × centre + b	× a	× a²	changes

The slide's example: x = 1, 2, 3, 4 has mean 2.5, median 2.5, SD 1.29, IQR 2. Then x + 5 has 7.5, 7.5, 1.29, 2, and 2x has 5, 5, 2.58, 4. (The slide's "compare x to z" in case one should read "x to y".) The variance and CV columns are extensions you can derive from the rules; they appear in textbook HW 43–46 and 52.

الشرح بالعربي — التحويل الخطي

إذا **أضفنا** رقمًا ثابتًا لكل القيم (مثل 5 درجات إضافية لكل الطلاب): المتوسط والوسيط يزيدان بنفس المقدار، أما الانحراف المعياري والمدى الربيعي **فلا يتغيران**، لأن البيانات انزاحت كلها دون أن تتباعد. إذا **ضربنا** كل القيم في ثابت: كل المقاييس تُضرب فيه، **والتباين يُضرب في مربعه**. وتحويل `°F = 1.8°C + 32` مثال على الحالتين معًا.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · **02 · Lesson 3-2: Measures of Variation** · 03 · Lesson 3-3: Measures of Position · 04 · Lesson 3-4: Exploratory Data Analysis · 05 · Revision pack: formulas, decisions, traps, R · 06 · Exam-style questions (no answers) · 07 · Full worked answers